

A Stochastic Resource-Extraction Economy of Earth

Geometry, Uncertainty, Extraction Economics, and Interdisciplinary Extensions

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Abstract

This paper develops a stochastic resource-extraction economy for a spherical planet Earth. Earth is modelled as a sphere with uncertain radius and mass. Radius and mass are assumed to follow log-normal distributions while extraction prices and costs follow normal distributions. Closed-form expressions are derived for extractable mass, expected revenue, expected cost, and expected profit. The framework is extended to economics, finance, welfare economics, warfare economics, international relations, political economy, data science, machine learning, and planetary resource management. Several theoretical propositions are established and geometric visualizations are provided using PGF/TikZ.

The paper ends with “The End”

Contents

1 Introduction	1
2 Geometric Foundations	1
3 Stochastic Earth	2
3.1 Mass Distribution	2
3.2 Radius Distribution	2
4 Density Distribution	2
5 Excavation Geometry	3
6 Price and Cost Uncertainty	4
7 Revenue, Cost and Profit	4
8 Expected Profit	4
9 A Fundamental Earth Resource Equation	5
10 Economic Interpretation	5
11 Finance Interpretation	5
12 Welfare Economics	5
13 Political Economy	6
14 International Relations	6
15 Warfare Economics	6
16 Psychology	6
17 Psychiatry	7

18 Biology	7
19 Chemistry	7
20 Physics	7
21 Data Science	8
22 Artificial Intelligence	8
23 Computer Science	8
23.1 Monte Carlo Algorithm	8
24 Statistical Properties	9
25 Geometric Illustration	9
26 Conclusion	9

List of Figures

1 Geometric Illustration.	9
-----------------------------------	---

List of Tables

1 Distributional assumptions.	9
---------------------------------------	---

1 Introduction

Resource extraction has historically been a central component of economic development, state formation, industrialization, military capability, and technological progress.

At planetary scales, uncertainty exists regarding:

- total resource stocks,
- extraction costs,
- market prices,
- technological feasibility,
- political constraints,
- environmental impacts.

This paper constructs a stochastic model beginning from elementary geometry and extending toward a generalized planetary resource economy.

2 Geometric Foundations

Assume Earth is a sphere.

Let

- $R > 0$ denote planetary radius,
- $M > 0$ denote planetary mass,
- $\rho > 0$ denote uniform density.

The volume of Earth is

$$V = \frac{4}{3}\pi R^3.$$

Hence density satisfies

$$\rho = \frac{M}{V}.$$

Substituting for volume,

$$\rho = \frac{3M}{4\pi R^3}.$$

3 Stochastic Earth

3.1 Mass Distribution

Assume

$$M \sim \text{LogNormal}(\mu_M, \sigma_M^2).$$

Equivalently,

$$\ln M \sim N(\mu_M, \sigma_M^2).$$

The moments are

$$E[M] = e^{\mu_M + \frac{1}{2}\sigma_M^2},$$

$$\text{Var}(M) = (e^{\sigma_M^2} - 1) e^{2\mu_M + \sigma_M^2}.$$

3.2 Radius Distribution

Assume

$$R \sim \text{LogNormal}(\mu_R, \sigma_R^2).$$

Thus

$$\ln R \sim N(\mu_R, \sigma_R^2).$$

and

$$E[R] = e^{\mu_R + \frac{1}{2}\sigma_R^2}.$$

4 Density Distribution

Since

$$\rho = \frac{3M}{4\pi R^3},$$

taking logarithms gives

$$\ln \rho = \ln \left(\frac{3}{4\pi} \right) + \ln M - 3 \ln R.$$

Proposition 1. *If M and R are independent log-normal random variables then ρ is log-normal.*

Proof. A linear combination of independent normal random variables is normal.
Since

$$\ln M \sim N(\mu_M, \sigma_M^2)$$

and

$$\ln R \sim N(\mu_R, \sigma_R^2),$$

it follows that

$$\ln \rho$$

is normal.

Therefore ρ is log-normal. □

The resulting parameters are

$$\mu_\rho = \ln\left(\frac{3}{4\pi}\right) + \mu_M - 3\mu_R,$$

$$\sigma_\rho^2 = \sigma_M^2 + 9\sigma_R^2.$$

5 Excavation Geometry

Suppose excavation is feasible up to radius

$$r_{\max} \leq R.$$

The infinitesimal shell volume is

$$dV = 4\pi r^2 dr.$$

Hence

$$dM = 4\pi \rho r^2 dr.$$

Integrating from the centre to the accessible radius gives

$$M_{\text{ext}} = \int_0^{r_{\max}} 4\pi \rho r^2 dr.$$

Thus

$$M_{\text{ext}} = \frac{4\pi}{3} \rho r_{\max}^3.$$

Substituting density,

$$M_{\text{ext}} = M \left(\frac{r_{\max}}{R} \right)^3.$$

Theorem 1 (Extractable Mass Theorem). *The fraction of planetary mass accessible for extraction equals*

$$\left(\frac{r_{\max}}{R} \right)^3.$$

Proof. Immediate from volume scaling of spheres. □

6 Price and Cost Uncertainty

Let extraction price satisfy

$$p \sim N(\mu_p, \sigma_p^2).$$

Let extraction cost satisfy

$$c \sim N(\mu_c, \sigma_c^2).$$

Define margin

$$m = p - c.$$

Then

$$m \sim N(\mu_p - \mu_c, \sigma_p^2 + \sigma_c^2).$$

The profitability probability becomes

$$P(m > 0) = \Phi\left(\frac{\mu_p - \mu_c}{\sqrt{\sigma_p^2 + \sigma_c^2}}\right).$$

7 Revenue, Cost and Profit

Revenue:

$$\mathcal{R} = pM\left(\frac{r_{\max}}{R}\right)^3.$$

Cost:

$$\mathcal{C} = cM\left(\frac{r_{\max}}{R}\right)^3.$$

Profit:

$$\Pi = (p - c)M\left(\frac{r_{\max}}{R}\right)^3.$$

8 Expected Profit

Using independence,

$$E[\Pi] = E[p - c]E[M]E[R^{-3}]r_{\max}^3.$$

Since

$$E[p - c] = \mu_p - \mu_c,$$

and

$$E[R^{-3}] = e^{-3\mu_R + \frac{9}{2}\sigma_R^2},$$

we obtain

$$E[\Pi] = (\mu_p - \mu_c)e^{\mu_M + \frac{1}{2}\sigma_M^2}e^{-3\mu_R + \frac{9}{2}\sigma_R^2}r_{\max}^3.$$

Therefore

$$E[\Pi] = (\mu_p - \mu_c) \exp\left(\mu_M - 3\mu_R + \frac{1}{2}\sigma_M^2 + \frac{9}{2}\sigma_R^2\right)r_{\max}^3.$$

9 A Fundamental Earth Resource Equation

Define accessibility

$$\kappa = \left(\frac{r_{\max}}{R} \right)^3.$$

Then

$$0 \leq \kappa \leq 1.$$

The entire economy reduces to

$$\Pi = (p - c)\kappa M.$$

10 Economic Interpretation

The model contains three multiplicative components:

1. Resource stock M ,
2. Accessibility κ ,
3. Net margin $(p - c)$.

A nation can become wealthier through:

- discovering larger resource stocks,
- improving extraction technology,
- increasing commodity prices,
- lowering extraction costs.

11 Finance Interpretation

The profit expression resembles an option valuation problem.

Define

$$X = \kappa M.$$

Then

$$\Pi = mX.$$

This resembles a leveraged claim on a stochastic underlying asset.

Resource companies can therefore be viewed as derivative contracts written on geological reserves.

12 Welfare Economics

Suppose social welfare is

$$W = U(C) - D(E),$$

where

- C denotes consumption,
- E denotes environmental damage.

The social planner solves

$$\max_{r_{\max}} W.$$

Thus extraction may be privately profitable but socially suboptimal.

13 Political Economy

Control of extractable resources creates:

- rents,
- taxation capacity,
- state power,
- political influence.

Resource-rich states often experience strategic competition over ownership rights.

14 International Relations

Countries possessing large expected values of

$$E[\Pi]$$

become strategically important.
Resource endowments influence:

- alliances,
- sanctions,
- trade agreements,
- security guarantees.

15 Warfare Economics

Suppose military strength satisfies

$$S = \alpha + \beta\Pi.$$

Then

$$\frac{\partial S}{\partial \Pi} = \beta.$$

Resource extraction directly finances military capability.

Historically, coal, oil, iron, copper, uranium, and rare earth elements have all contributed to strategic power.

16 Psychology

Humans systematically overestimate resource abundance during booms and underestimate scarcity during crises.

Behavioral biases include:

- optimism bias,
- overconfidence,
- recency bias,
- herd behaviour.

These biases affect both p and c .

17 Psychiatry

Resource booms and collapses influence:

- stress,
- anxiety,
- migration,
- social cohesion.

Extreme commodity shocks have historically contributed to mental health crises.

18 Biology

Life ultimately depends on resource extraction.

Examples include:

- nutrient extraction from soil,
- mineral cycling,
- energy acquisition.

Biological systems themselves may be interpreted as resource-allocation mechanisms.

19 Chemistry

Extraction economics is fundamentally constrained by chemistry.

Examples:

- ore grades,
- reaction yields,
- separation efficiencies,
- energy requirements.

Thermodynamic limits impose lower bounds on extraction costs.

20 Physics

Mass-energy equivalence implies

$$E = Mc^2.$$

If sufficiently advanced technology existed, the ultimate energetic value of extractable mass would be

$$E_{\text{ext}} = M_{\text{ext}} c^2.$$

Thus

$$E_{\text{ext}} = M \left(\frac{r_{\text{max}}}{R} \right)^3 c^2.$$

21 Data Science

The model naturally suggests estimation using:

- Bayesian inference,
- Monte Carlo simulation,
- particle filtering,
- state-space models.

Parameter vector:

$$\theta = (\mu_M, \sigma_M, \mu_R, \sigma_R, \mu_p, \sigma_p, \mu_c, \sigma_c).$$

22 Artificial Intelligence

AI systems can optimize extraction through:

- geological prediction,
- drilling optimization,
- autonomous mining,
- commodity forecasting.

A reinforcement-learning formulation is

$$\max_{\pi} E \left[\sum_{t=0}^{\infty} \gamma^t \Pi_t \right].$$

23 Computer Science

23.1 Monte Carlo Algorithm

Input N

For i=1,...,N

Draw M_i

Draw R_i

Draw p_i

Draw c_i

Compute

Pi_i = (p_i - c_i) M_i (r_max/R_i)^3

End

Estimate:

Mean(Pi_i)

Variance(Pi_i)

Quantiles(Pi_i)

24 Statistical Properties

Variable	Distribution
M	Log-normal
R	Log-normal
ρ	Log-normal
p	Normal
c	Normal
$m = p - c$	Normal
Π	Product-mixture distribution

Table 1: Distributional assumptions.

25 Geometric Illustration

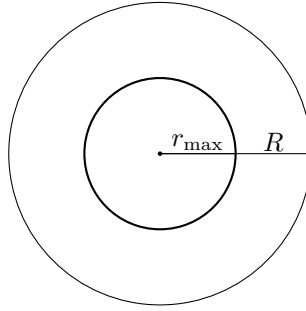


Figure 1: Geometric Illustration.

The inner sphere denotes the accessible extraction region.

26 Conclusion

Starting from elementary geometry, we derived a stochastic planetary extraction economy. The resulting model combines geometry, probability theory, economics, finance, political economy, warfare economics, international relations, physics, chemistry, biology, psychology, psychiatry, computer science, data science, and artificial intelligence.

The fundamental equation

$$\Pi = (p - c)M \left(\frac{r_{\max}}{R} \right)^3$$

captures the interaction of resource abundance, accessibility, and profitability under uncertainty.

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Glossary

M	Planetary mass.
R	Planetary radius.
ρ	Planetary density.
$r_{\max}$	Maximum accessible excavation radius.
p	Unit extraction price.
c	Unit extraction cost.
m	Profit margin.
Π	Profit.
$\mathcal{R}$	Revenue.
$\mathcal{C}$	Cost.
κ	Accessibility coefficient.
W	Social welfare.
S	Military strength.
θ	Model parameter vector.

The End